

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 9

SCALE OF PRODUCTION AND LAWS OF RETURN

MEANING OF SCALE OF PRODUCTION

A firm or an entrepreneur have to face so many problems, but the most important problem is selection of means of production or scale of production, as the profit and loss of the whole business depends on it. Any good can be produced by the following two means.

1ST :

To produce the good by less capital with the help of simple instruments. it is called small scale production. e.g. a shoemaker making shoes, a barber rendering services etc.

2ND :

To produce the good by large amount of capital with the bulk of raw material and modern machines. It is called large scale production. e.g. production of cloth in a textile mill, production of shoes in shoe factory etc.

The following factors determine the scale of production which includes the size of business, combination of inputs used and volume of production.

1. QUANTITY OF CAPITAL:

If the entrepreneur is financially stable and has an easy access to modern machines and a bulk amount of raw material then he could start his business on large scale and if he is in lack of Capital, then goods will be produced on small scale.

2. DEMAND FOR THE GOODS:

When the goods have a greater demand and have a vast market will be profitable on the other hand if, the good has low demand, then it will be produced on small scale.

3. TECHNICAL POSSIBILITIES:

Technical Possibilities should be kept in view while determining the scale of production. Modern and automatic machines are used in the processing of some goods e.g. Sugar, paper, cement etc. but some goods are produced only on small scale due to technical possibilities e.g. production of jewelry, furniture etc.

In this context the most important phenomena is skilled labour, if labour is skilled the good could be produced on large scale otherwise, small scale would be better.

4. MEANS OF TRANSPORTATION:

Developed means of transportation e.g. Roads, rail, sea routes, Air Cargo etc., encourage large scale production because extent of markets expands, on the other hand due to

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un-developed transportation means the goods will be produced on small scale.

5. NATURE OF DEMAND:

if the demand for good is elastic then it would be produced on large scale. As the demand can be increased many times by making a small decrease in price. But if the demand for good is less elastic then it would be produced on small scale.

6. FISCAL POLICY:

if the fiscal policy is in the favour of entrepreneurs and they have given the facility of subsidies and Govt. encourages privatization, then it is profitable to produce the goods on large scale otherwise, it would be better to produce the goods on small scale.

7. POWER RESOURCES:

Power resources like stream, electricity, oil and gas determine the scale of production. If these are available easily to the entrepreneur, the entrepreneur could produce the goods on large scale.

8. ENTREPRENEURIAL ABILITY:

if the entrepreneurs are skilled and experienced, the goods will be produced on large scale, otherwise small scale would be better to produce the goods.

9. POLITICAL STABILITY:

if a country is politically stable, business can expand on large scale but in case of unstable political circumstances business will be limited to small scale production.

INTERNAL AND EXTERNAL ECONOMIES

An entrepreneur can get the following economies while producing the good on large scale.

A. INTERNAL ECONOMIES:

These economies are related with the internal circumstances of the business e.g if an entrepreneur produces a good at a very low cost due to his skill and ability.

Internal economies can be achieved in the following forms.

1. ADMINISTRATIVE ECONOMIES:

Under large scale production, more efficient and skilled labour is used but the number of labourers is limited in this way cost is reduced and internal economies are achieved.

2. COMMERCIAL ECONOMIES:

In large scale production, the raw material is purchased in a bulk quantity which is available at low prices. In this way total expenditure is reduced.

3. FINANCIAL ECONOMIES:

While, producing the goods at large scale, the firm can easily get loans from banks. In this way more finance can be available to the business.

4. BY-PRODUCTS ECONOMIES:

In large scale production, the rest of raw material can be used in by-products e.g the low quantity thread which rest from textile firm can be used in formation of mats etc.

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5. ECONOMIES IN THE USE OF FACTORS:

The best combination of factors of production can be achieved which proves very helpful in lowering the cost.

B. EXTERNAL ECONOMIES:

These economies are not specifically related to a firm. All firms and institutions are equally benefited with these economies.

1. TRANSPORTATION FACILITIES:

If many a firms start producing the same goods in the same area, then Govt. gives the facility of transportation. New roads and railway lines are came into being. And it is easy for the entrepreneur to move from one market to other market.

2. FACILITIES OF BANKING AND INSURANCE COMPANIES:

Banks always like to open their branches in commercial areas. In this way it is easy to get the loan for enterprenures. Business men get the chance of Insurance by the establishment of Insurance Companies.

3. REPAIRING FACILITIES:

Workshopes and repairing centers are established near the factories. In this way any fault in the machines can be immediately removed without the wastage of time and resources.

4. RESEARCH CENTERS:

Research Centers and labouratories are established. Due to which modern and innovated means of production are invented. And all the eterpreneures can be facilitated.

5. AVAILABILITY OF TECHNICAL SKILL:

When a large number of institutions and factories work at the same place, then people start getting experience and skills. In this way the demand for skilled labour can be easily fulfilled.

DIS-ECONOMIES

When a firm's scale of production is greater than the optimal scale, then it's economies are abolished and there start dis-economies. In this way, the firm's average cost start increasing and profit tends to decrease.

Dis-Economies of business are given as Under.

1. ADMINISTRATIVE PROBLEMS:

When a firm's scale of production is so vast then many sort of administrative problems come into being. Due to which the collective progress of the firm is effected e.g

- (a) There exists a communication gap between entrepreneur and labour. So it is difficult for the enterpreneure to keep an eye on all the labourers.
- (b) It is difficult to control the labour activities like strikes, street march etc.
- (c) It is difficult to keep in touch with different business departments.
- (d) There is a big gap between policy making and it's implementation. In this way, the time is wasted.

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2. UN-NECESSARY COMPETITION:

When a business goes on expanding, many firms enter in the industry. There starts a competition to attain the required quantity of factor of production. The firms want to get the factors of production even at a very high price. In this way the average cost is increased.

3. TRANSPORTATION PROBLEMS:

Availability of raw material and transportation of finished goods to the market is a great problem for the large size firms. They have to pay more for transportation. In this way, cost is increased.

4. INCREASE IN WAGES:

Demand for labour increases due to expansion in business. This increase in demand cause an increase in wage rate. In this way, cost of production goes up.

5. SCARCITY OF RAW MATERIAL:

Demand for raw material increases due to increase in the size of business. If demand for raw material is more than it's supply, it will cause an increase in price which will push the cost upward.

6. DECREASE IN RETURN:

Sometimes output of firms increases too much which pulls pressure on it to lower down it's prices. Hence the firm's revenue is decreased, on the other hand there starts a price war among firms which may create a great disturbance in business world.

7. FAVORITISM:

Ineligible persons are employed due to expansion of business which will directly effect, the total output and cost and production.

MERITS OF LARGE SCALE PRODUCTION:

When production have been done on large scale and machinery is used for production and So many workers are involved in the production, it is called large scale production.

There are following characteristics of large scale production :

1. DIVISION OF LABOUR:

Division of labour is adopted in large scale production. In this way, the same work is done by a person again and again which increases his ability and skill, consequently firm's efficiency is increased.

2. SERVICES OF TRAINED SKILL AND TECHNICAL KNOWLEDGE:

When production is on large scale, firm has an opportunity to hire technically trained staff which increases it's efficiency.

3. ECONOMIES IN PURCHASE AND SALE:

Raw Material is purchased from the wholesale markets. It causes purchase economies. Final goods are sold in high price markets. In this way profit increases.

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4. FIXED COST DECREASES:

Fixed cost of a firm also decreases due to large scale production.

5. RESEARCH AND TRAINING CENTERS:

Research and training centers can be established with the help of widely available resources under large scale production. In this way modern means of production are invented and utilized.

6. REPAIRING CENTERS:

As there have been available the facility of workshops and training centers near the firms. It can save the time and resources.

7. USE OF BY-PRODUCTS:

Wastage of large scale production firms can be used by the firm in the Production for process By-products.

8. ADVERTISEMENT:

Publicity can be carried out through advertisement. Which is an effective tool to increase the demand for the good.

9. EMPLOYMENT OPPORTUNITIES:

Employment rate is increased due to large scale production. In this way the living standard of the people can be improved.

10. INCREASE IN GOVT. REVENUE:

Govt. revenue in the form of taxes increases due to large quantities of production.

11. SUBSIDIES AND TAX CONCESSION:

Govt. has given subsidies and tax concession to the firms engaged with large scale production.

12. POWER TO FACE LOSS:

As the number of share holders is greater in large scale production. So they have the power to face loss.

13. MORAL VALUES:

In large scale of Production, labour have a chance to improve morality. They learn norms and ethics which is good for the firm and society.

14. FOREIGN EXCHANGE EARNING:

Large scale production helps in earning foreign exchange due to export of manufactured goods.

15. INTERNAL AND EXTERNAL ECONOMIES:

Large scale production is helpful to achieve the internal and external economies, in this way, there is an expansion in the business.

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DEMERITS OF LARGE SCALE PRODUCTION

There are only a few demerits of Large scale Production given as under :

1. LACK OF PERSONAL CONTACTS:

Thousands and millions of people are engaged in large scale production. Due to which mutual understanding can not exist between labour and Enterprenure because they are not in direct contact with each other. That's why most of misunderstandings are created while sometimes this situation leads to strikes. This has a bad impact on production.

2. DANGER OF OVERPRODUCTION AND DEPRESSION:

Mostly the estimates about the demand of goods are proved wrong, the goods are not totally sold. The profits of investors decrease and their investment becomes limited. In this way unemployment increases and the economy becomes the victim of depression.

3. ESTABLISHMENT OF MONOPLIES:

The established enterprenure may form monopoly. The small businessman can't compete them. Sometimes they have to leave their business. In this way the monopolists charge the price of their own choice and bother the consumer's.

4. LACK OF CREATIVE ABILITIES:

If a man has to do the same kind of work, then he becomes victim of boredom and does not take interest in his work.

5. TENSION AND TIRESOME:

Due to working on machines the labourers become tired and their health diminishes day by day. This will effect their efficiency of work.

6. ACCIDENTS:

The labourers working on big machines have to face accidentens and sometime, they have to loose their lives.

7. SOCIAL EVILS:

In many factories, men women and children work together, in this way many social evil's can be spread. Like smoking , drinking, etc.

8. LACK OF PERSONAL CHOICE:

The decision of goods produced on large scale is done by enterprenure. In this way the consumer's personal choice is neglected.

9. FOREIGN DEPENDENCE:

We have to depend on foreign countries for finance, machinery and raw material. If due to any reason these things can't be imported, then the business will be effected.

10. DIS-ECONOMIES:

if the business is extended up to a certain limit. Then the producing unit has to face diseconomies resulting a decrease in profits.

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SMALL SCALE PRODUCTION

MERITS

1. PERSONAL INTEREST AND CONTROL:

If the business is limited, then enterpreneur can keep an eye on it easily and enjoy more profit with personal interest.

2. CONSUMER'S CHOICE:

In small scale business consumer's choice is preferred and goods are prepared on the order of consumer due to this over production cannot occur.

3. BETTER LABOUR RELATIONSHIP:

As in small scale production only a few workers are employed that's why there is a direct relationship between workers and enterpreneurs. They understand the problems of each other. In this way they become well wisher's of each other. Co-operation is increased between them. Thus production is increased.

4. SMALL SCALE BUSINESS:

As the small scale business can be started with little capital, so a poor man can also start business and earn his livli-hood.

5. WORK NEAR RESIDENCES:

As the small scale industry is situated near houses so all the family members can easily participate in it. Moreover, children and other workers can be easily treated.

6. QUICK DECISION:

If the business man has to take the decision about his business. Then his work will not suffer from wastage of time because he has not to fulfill any official formality.

7. ECONOMY IN ADMINISTRATIVE EXPENDITURE:

In small scale business all the arrangements should be managed by enterpreneur In this way more staff is not needed. And expenditures can be reduced.

8. SOCIAL EVILS ERADICATION:

In small scale industry more people do not meet each other. In this way social evils are avoided.

9. SAFE FROM ACCIDENTS:

In small scale industry, heavy machinery is not used, in this way accidents are reduced.

10. NO DANGER OF OVER PRODUCTION:

In small scale production the business is limited. Mostly things are produced on order. In this way there is no danger of over production.

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DEMERITS

1. NO INTERNAL AND EXTERNAL ECONOMIES:

As in small scale production business is limited so internal and external economies can not be achieved.

2. FINANCIAL WEAKNESS:

In small scale business, there is lack of capital. So the business man has to depend on old techniques of production. In this way the production becomes limited.

3. DIFFICULTY IN OBTAINING LOANS:

As the owner of small scale business is not financially stable. He has nothing to offer as guarantee so, the banks refuse to give them loan.

4. LACK OF RESEARCH AND EXPERIMENT:

As research work requires more capital and small scale business men lack capital so they can not afford new research and modern methods.

5. LACK OF POWER TO COMPETE AND FACE LOSS:

If the small scale business man have to face loss. It becomes difficult for him as that is beyond his financial power. In this way the business has the danger of shut down.

6. HIGH PRICES OF RAW MATERIALS:

As in small scale industry the raw material is used in small quantity. So the businessman has to purchase it at market price, which is comparatively costly. Thus the cost of production is increased.

7. NO BY-PRODUCTS:

As in small scale business there is no use of extra material because they have no arrangements to produce by-products. In this way extra material will be wasted.

8. WASTAGE OF TIME:

As division of work is not possible. Thus only one person have to do each and every part of work. In this way lot of time will be wasted. There is not saving of tools and if tools become out of order then no work shop is available near by for this purpose, a business man have to come in big cities. In this way finance as well as time is wasted.

9. NO DIVISION OF LABOUR:

In small scale division of work is not possible. In this way the business can't flourish and the cost of production can't be decreased.

10. DIFFICULTY IN THE USE OF MODERN MACHINES:

If the business is on small scale then the use of heavy machinery is not possible. So primitive tools are used which have high cost of production.

LAWS OF RETURN

If a business has been started and more units of factors of production are added into it for the scale of expansion, then it passes through three phases:

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1. Every additional unit gives more production to a certain limit. It means, marginal production increases gradually.
2. Every additional unit gives production which is equal to that of first unit. It means marginal production remains constant.
3. Every additional unit gives production which is less than that of first unit. It means marginal production decreases.

THE ABOVE SITUATIONS REPRESENT LAWS OF RETURN AS UNDER :

1. Law of Increasing Return.
2. Law of Constant Return.
3. Law of Decreasing Return.

LAW OF INCREASING RETURN

If Marginal production increases due to increase in the units of a variable factor of production in any business. Then this business obey the law of increasing return. This law can be defined as Under.

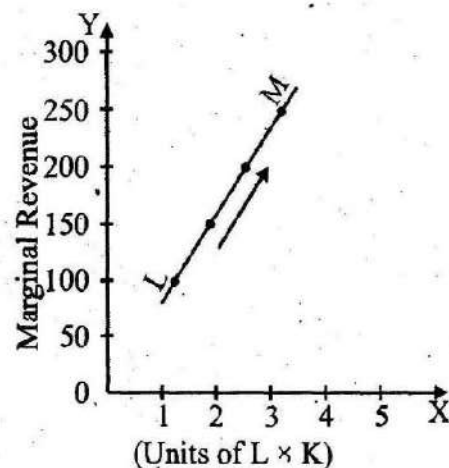
“In any production process if the marginal productivity of variable factor, when it is combined with fixed factor, is increasing then this situation leads to the law of increasing returns”.

In the following schedule. It is assumed that land is a fixed factor of production. Its's quantity is 10 Acre (which never changes) but labour and capital are variable factors. So when we put an additional unit of these variable factors in a business then marginal production increases gradually to a certain extant.

“SCHEDULE”

Units of Labour & Capital	Total Revenue	Marginal Revenue
1	100	100
2	250	150
3	450	200
4	700	250
5	1000	300

This schedule explains that as we increase the units of labour and capital, the marginal production increases gradually and the production increases more than before at every unit. We can also explain it with the help of a diagram



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Explanation:

We take the units of labour (L) and Capital (K) on x-axis and Marginal revenue on y-axis. On the basis of the data given in schedule, we draw the LM Curve. This curve has been drawn by joining the intersection points of labour and capital and marginal revenues. This is a positively sloped curve. Thus it is the curve of increasing returns and indicates that as we put more units of labour and capital in a business Marginal productivity increases gradually.

Assumption:

1. Fixed factor of production should neither be changed nor divisible.
2. The units of variable factors of production should be of same quality and standard.
3. The units of variable factors should be available as needed.
4. Methods of production should not be changed.

Keeping in view the above given assumptions, the law of increasing returns can be proved true.

APPLICATION OF THE LAW OF INCREASING RETURNS:

This law mostly applies in industrial sector, because this sector is handled by human being instead of nature and a man can handle his business under the law of increasing returns due to his experience, intelligence, competence for a long time. But this law is not always applicable on Industrial sector. When the business is extended and optimal level of production is attained the marginal production begins to decrease, after such as level this law will not be applicable on business.

LAW OF CONSTANT RETURN

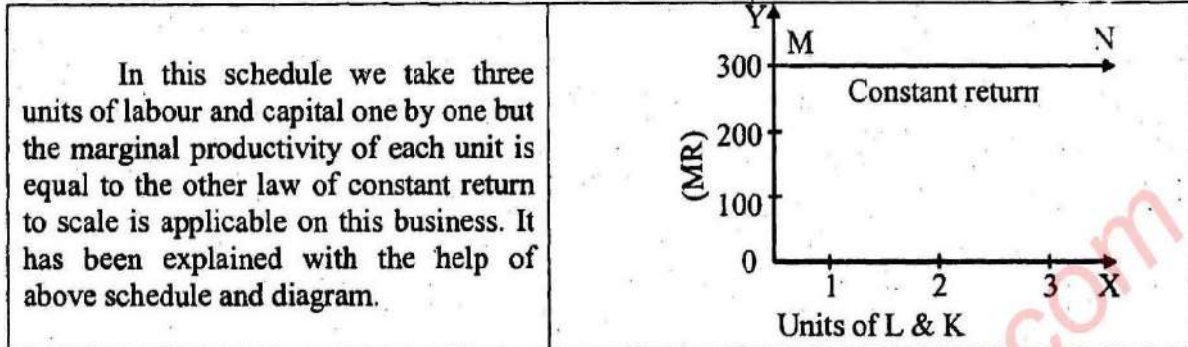
If, the marginal production remains constant, when we put more units in business. Then constant return to scale is applicable on business. It is defined as under :

"In any production process if the marginal productivity of variable factors when combined with fixed factor, remains constant. Then this situation is called law of constant return."

To explain this law, it is assumed that land is a fixed factor of production but the units of labour and capital can be changed and when marginal productivity remains constant by putting more units of variable factors. Then the law of constant return to scale is applicable on business. This law starts when the law of increasing returns to scale ends and works till the law of decreasing returns starts.

Units of L & K	TR	MR
1	300	300
2	600	300
3	900	300

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EXPLANATION:

In this diagram, we take the units of labour and capital on x-axis and marginal return on y-axis. If we match the values of labour and capital and marginal productivity according to schedule, we get the curve MN i.e. parallel to x-axis. Thus this is the curve of law of constant returns and explains that if we put more units of labour and capital into business there will be no change in Marginal productivity.

APPLICATION OF LAW OF CONSTANT RETURN:

This law is applicable on those industries which are equally effected by nature and human efforts e.g wool industry. Wool is obtained from sheeps which are effected by nature on the other hand man prepares blankets, shawls and other things with wool by his own skill.

Both forces compete each other and there is a balance between them. Hence, marginal production remains the same. It must be kept in mind that such industries do not always remains under the constant returns to scale. Ultimately these become under the law of "Decreasing Return".

LAW OF DECREASING RETURN

When in any business, one factor of production land is assumed to be fixed and we put more units of Capital and labour on it. Then to a certain limit every additional unit gives less productivity than the first unit.

This situation is called "Decreasing Returns to Scale".

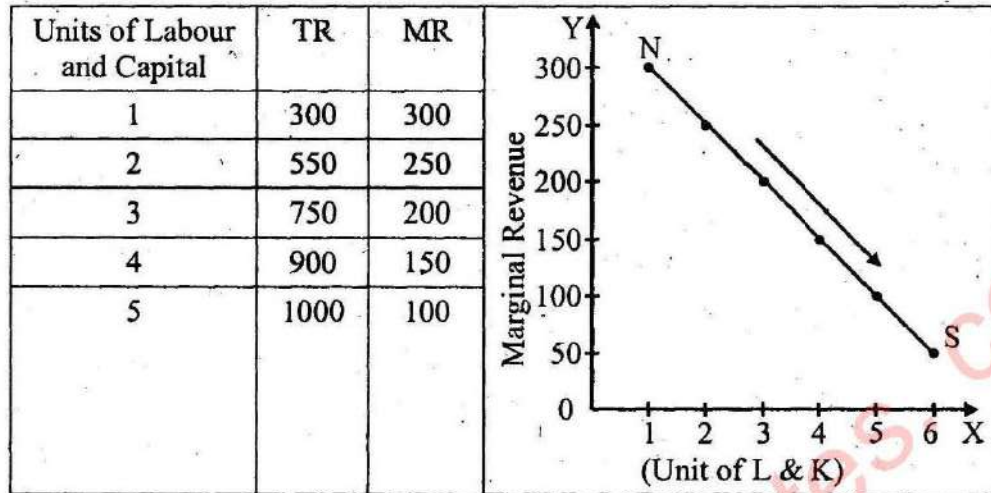
Economists define this law as under :

"In any production process if the marginal productivity of variable factors (L and K), when combined with fixed factor (N) is decreasing. Then this situation leads to decreasing return and is called law of decreasing return".

This law is applicable when after optimal level, the law of constant return becomes ineffective.

This law can be explained by the following schedule in which land is a fixed factor of production but the units of labour and capital are put one by one. Every additional unit gives less output than the previous unit.

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This schedule shows that as we put more units of labour and capital, the marginal revenue is going to be decreased. Hence the law of decreasing returns is applicable

EXPLANATION:

In this diagram x-axis represents the units of labour and capital and y-axis shows the marginal revenue. Diagram shows that the combination of the units of labour and capital and MR gives us NS Curve which is negatively sloped and shows that every additional unit gives less output than the previous unit and Marginal production tends to decrease.

ASSUMPTIONS:

1. No change should occur in fixed factor of production.
2. There should be no change in means of production.
3. The units of variable factors should be homogenous.
4. The level of optimal combination of factors of production should reach at an end.

APPLICATION OF THE LAW:

This law is applicable on every business. No matter the business is related to industrial sector or agriculture sector. But the sectors which are very close to the nature are directly and soon effected by this law because :

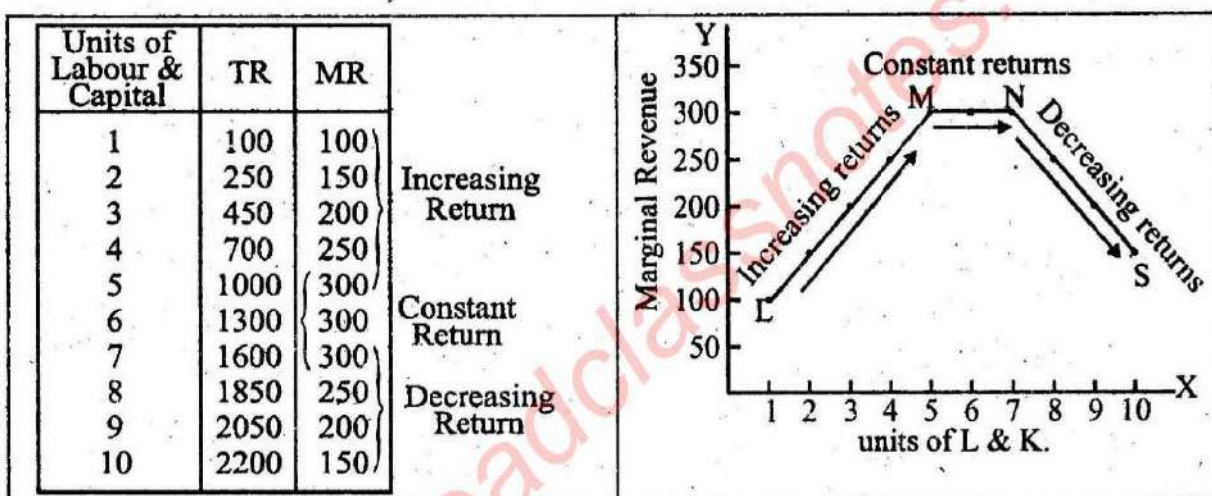
1. Natural disasters e.g floods, famines, droughts badly effect the agriculture output.
2. Land is limited and it's fertility can be increased only to some extent.
3. The principle of division of labour can be applicable to a certain level. When the business expands, expenditure also expands and output decreases.
4. In agriculture sector, out-dated instruments are used and the additional units give less output.

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"COMBINED REPRESENTATION OF THE LAWS OF RETURNS"

All above discussed laws of returns can be explained by the same schedules and diagrams while, combining them at one place.

Initially when a business is started and we put the units of labour and capital at this time the production process is under the law of increasing return. After an optimal limit every additional unit gives the same output. Here law of constant return is applicable when after a certain level output starts decreasing While introducing an additional unit of variable inputs law of decreasing return is applicable.



EXPLANATION:

In this schedule first five units show the increasing returns because there is an increase in Marginal production.

Units 5th to 7th show constant returns as there is no change in Marginal production.

After 7th unit the marginal production starts decreasing which shows the decreasing return.

In this diagram x-axis represents the units of labour and capital and y-axis represents the marginal revenues. LM is the curve of increasing return which is positively sloped it shows as we put more units of labour and capital. The marginal production tends to increase.

MN curve shows constant return and is parallel to x-axis which shows that by putting more units of capital and labour. There will be no change in the marginal production and it remains constant.

NS curve shows decreasing return which is positively sloped showing that by putting more units of labour and capital, the marginal production is going to be decreased.

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LAWS OF REVENUE AND THEIR RELATION WITH THE LAWS OF COST:

Laws of returns and laws of cost are applicable on the business at the same time. Because increasing returns may be given the name of decreasing cost. When the marginal return increases the marginal cost decreases.

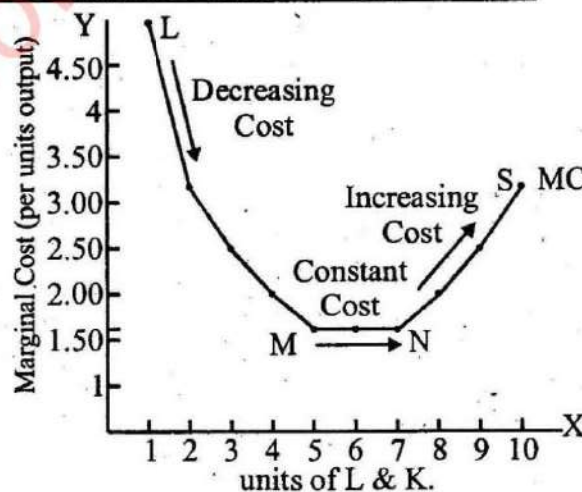
When there is constant returns, cost per unit of output remains the same which is regarded "Law of Constant cost".

When there is decreasing returns, there is an increase in the cost per unit. Which shows the increasing cost.

All these three laws of return can be shown by the following schedule.

Let us assume that the cost of a unit of labour and capital is Rs. 500/- and cost being constant at every unit. But when the Marginal production increases the cost of production per unit decreases. It remains constant when Marginal production is constant and increases when marginal production decreases.

Labour and Capital Units	TR	MR	MC (per unit output)	
1	100	100	5	Decreasing Cost
2	250	150	3.33	
3	450	200	2.50	
4	700	250	2.00	
5	1000	300	1.66	Constant Cost
6	1300	300	1.66	
7	1600	300	1.66	
8	1850	250	2.00	Increasing Cost
9	2050	200	2.50	
10	2200	150	3.33	



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In this schedule from 1st to 5th unit business is under decreasing cost as the Marginal returns go on increasing per Kg. But from 5th to 7th unit there will be no change in Marginal cost and hence the business is under constant returns or "Constant cost" after 7th unit, the Marginal cost per unit is going to be increased and the production is under decreasing returns or "increasing cost."

EXPLANATION OF DIAGRAM:

In this diagram x-axis shows the units of capital and labour and y-axis shows the Marginal cost per Kg. LM is the decreasing cost curve because it is a negatively sloped curve which shows that as we put more units of capital and labour, the Marginal cost per Kg is going to decrease NM Curve is parallel to x-axis and shows the constant cost because there will be no change in cost by putting the additional units of labour and capital. NS curve is positively sloped curve which shows that by putting more units of labour and capital cost per Kg is going to be increased.

Hence in business first decreasing cost then constant cost and ultimately, Increasing cost is applicable.

"LAW OF VARIABLE PROPORTIONS"

Modern economists have given the name :

"Law of variable proportion" or law of "non-proportional returns" to the above described laws of return as these conditions are the three phases of the same law.

This law of production explains that if we put the units of variable factor of production on the fixed factor of production, then every additional unit of variable factor increases the production more than before. But after a certain point, the proportion of this increment going to decrease consequently, the total production start decreasing after reaching it's peak.

In this way production process can be divided into three different phases.

"This law of production states that if variable quantity of one resource is applied to a fixed amount of other inputs, output per unit of variable input, will increase but beyond some point the resulting increase will be less and less, with total output reaching a maximum before it finally begins to decline".

PROF. W.J.L RYAN IN HIS BOOK "PRICE THEORY" WRITES:

"The law of Non-proportional returns states, with a given method of production the application of further, units of any variable input to a fixed combination of other factors will, until a certain point is reached, yield more than proportional increase in output and there after, less than proportional increase in output".

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“This law can be defined simply that :

“The marginal productivity of variable factor of production (L), combined with fixed factor of production land, (N) will eventually decline”.

“This law can be explained by following schedule. :

Units of Variable Factor	TP _L	AP _L	MP _L	Stages
1	20	20		1st Stage Increasing Return MP > AP → MP = AP
2	50	25	30	
3	90	30	40	
4	120	30	30	
5	140	28	20	2nd Stage MP < AP Decreasing Return
6	150	25	10	
7	154	22	4	
8	154	19.25	0	3rd Stage MP < 0 Negative Return
9	145	16.1	-9	
10	130	13	-15	

EXPLANATION:

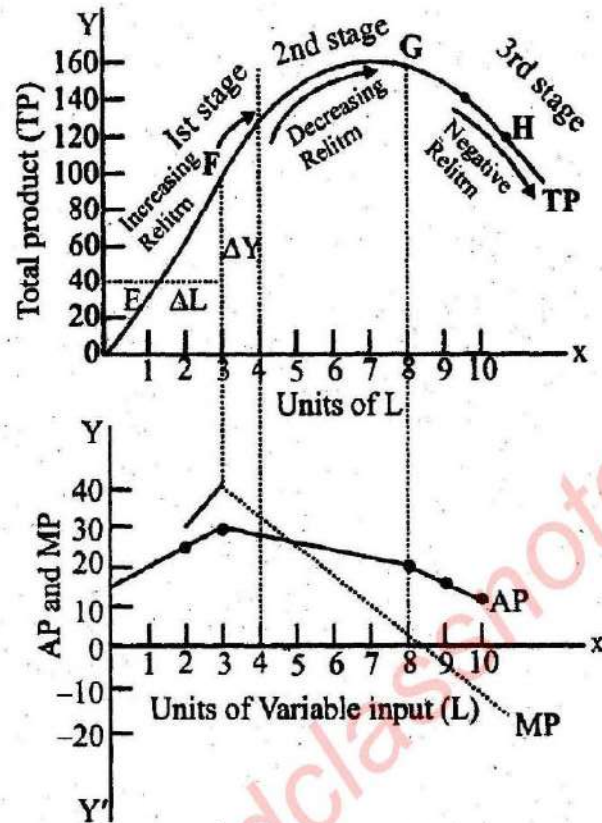
In this schedule the marginal production of variable factor increases upto 3rd unit and production process is under “increasing returns”. But at 4th unit average production does not decrease even with a decrease in marginal production and these both are equal. Which means that the fixed factor is more efficient as compared to variable proportion.

From 4th to 7th unit the proportion in increase in total production is going to be decreased and the total production of 7th and 8th unit is equal and the marginal production tends to decrease and becomes “Zero” at 8th unit. This is the 2nd stage which shows “decreasing returns”.

After 8th unit, the total production tends to decrease and the marginal return is negative. Hence it is the 3rd stage of production process which shows negative returns.

“This can be shown by the following Diagram.

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EXPLANATION:

In this diagram x-axis shows the units of variable factor and y-axis shows the production. When the total production increases with an increasing ratio then marginal production increases upto 3rd unit and average production increases till the 4th unit and Hence it is the first stage of law of variable proportion.

In diagram EF part of TP curve explains this process. At this stage TP is going rapidly upwards. Marginal production tends to decrease from 4th to 7th unit and becomes 'Zero' at 8th unit. So the increment process of production also start decreasing and Hence it is 2nd stage and business is under "decreasing returns". In the diagram FG part of TP curve explains it. At this stage total production reaches at its peak and Hence this curve slowly goes upwards as after point F. the marginal production is going to decrease, hence this point is called point of "inflexion". 3rd stage is negative returns. The total production is going to decrease and marginal production becomes negative in this stage. There starts negatively returns after 8th unit in the schedule, while in diagram GH part of TP curve shows this stage which is downwardly sloping. The above mentioned two parts diagram. "TP" is total product curve "AP" is Average production curve and "MP" Marginal production curve. The mutual relationship between these curves is given as under :

1. When average production is increasing, the marginal production is greater than average production

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MP > AP

2. When average production is at its maximum level, it will be equal to marginal production.

MP = AP

3. When average production is diminishing, marginal production becomes less than it.

MP < AP

4. When TP starts declining MP becomes negative.

MP < 0

OBJECTIVE QUESTIONS

1. Select and Tick (✓) the correct answer;

1. "Law of diminishing return" is also known "Universal Law" Because?
 - a. It is applicable on every walk of life.
 - b. Every sector of economy is going to face this law.
 - c. It is the fruit of struggle of all entrepreneurs.
2. Law of diminishing return is mostly applicable on :
 - a. Communication. b. International trade. c. Industry. d. Agriculture.
3. If a business involves both the human labour and nature then which law is applicable on it :
 - a. Law of diminishing returns. b. Law of constant returns.
 - c. Law of increasing returns. d. No Law.
4. Point out the small scale production from the following
 - a. Shoemaker makes shoe. b. Service shoe factory.
 - c. Sugar mill. d. Textile mill.
5. What is the 3rd stage of law of variable proportion.
 - a. Positive returns b. Negative returns
 - c. Diminishing returns d. Increasing returns
6. When the business is under constant return, then what law of cost is applied :
 - a. Increasing cost b. Constant cost
 - c. Implicit cost d. Diminishing cost
7. When the business is under increasing return, then what law of cost is applicable :
 - a. Fixed cost b. Diminishing cost
 - c. Increasing cost d. Opportunity cost
8. From the following, what are not included in External economies.
 - a. The facilities of telecommunication
 - b. Establishment of research centers.
 - c. Financial economies
 - d. Establishment of banks and insurance companies.
9. Embroidery and sewing cloth by tailor is called :
 - a. Large scale b. Small Scale
 - c. Scale of return d. Scale of proportion.
10. When business expands beyond its optimum level then which type of economies are obtained :
 - a. Internal b. External c. Personal d. Non-economies.
11. Who presented the concept of economies of scale :
 - a. Alfred Marshall b. J.B Clark
 - c. Ricardo d. Pigou.

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12. When a variable factor is continuously combined with fixed factor and MP of each additional factor is constant then the production process is under :
- a. Increasing Returns b. Decreasing Return
c. Constant Returns d. Negative Return.
13. Law of Decreasing return is applied after a longer time on which sector :
- a. Agriculture b. Dairy farming
c. Poltery farming d. Industry
14. By-products are received under which
- a. Small scale b. Internal economies
c. Large scale d. External economies
15. Large scale is harmful due to:
- a. over production b. Power of Competition
c. Advertisement d. By-products.

"DESCRIPTIVE PART I"

1. What is meant by scale of production?

An important decision about production is to determine scale. Whether, it should be at large scale with bulk of capital or small scale with less capital.

2. Describe three internal economies of large scale production?

- 1) Administrative Economies
- 2) Commercial Economies
- 3) Financial Economies (By- products economics in use of factors economies.)

3. Give merits of large scale production?

- 1) Division of labour
- 2) Fixed cost decreases
- 3) Trained skilled staff
- 4) Research & training centers
- 5) Subsidies, tax concession etc.

4. Write names of law of returns?

- 1) Law of increasing return
- 2) Law of decreasing return
- 3) Law of constant return

5. What are the demerits of large scale production?

- 1) Lack of personal contacts
- 2) Danger of over production and depression
- 3) Establishment of monopolies etc.

6. Why does law of increasing return is called the law of diminishing cost?

In law of increasing return marginal Product/Revenue increases, and marginal cost decreases so at the same time it is applicable on business or production process. Thus it is called law of diminishing cost.

7. Write down five merits of small scale production?

- 1) Personal Interest and control
- 2) Consumers choice
- 3) Better relationship
- 4) Quick decision
- 5) Economy in administrative expenditure

8. Write down the stages of law of variable proportion?

- 1) Increasing return
- 2) Decreasing return
- 3) Negative Return

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9. What is the relationship between total production and marginal production?
When TP increases at increasing rate MP also keep on increasing rate. At 2nd stage TP increases at diminishing rate MP start decreasing. At 3rd stage when TP Start decreasing MP becomes negative.

10. What is the relationship between law of diminishing returns with law of increasing cost?
When there is decrease in returns, cost per unit increases that why it is named law of increasing cost.

11. Explain Internal and external economies?
The economies related with internal condition of business, depend upon skill and ability of producer which are called internal economies. Whereas external economies are not specifically related to any firm. All firms are equally benefited by these, such as; Electricity, Infrastructure banks etc.

12. What are dis- economics?
When a firm's scale is greater than the optimal scale then dis-economies start. In such a way firms average cost increases and profit decreases.

13. Define law of constant return?
In any production process if the marginal productivity of variable factors when combined with fixed factor, remains constant is called constant return. The return of every additional unit remains the same.

14. Define law of variable Proportions?
This Law states, if variable quantity of one resource is applied to fixed amount of other inputs, output per unit will increase but beyond a certain point, the resulting increase will be less with total output reaching maximum before it decreases.

15. Why law of decreasing return readily apply to agriculture sector?
The sectors which are closely related to the nature are directly and readily effected by this law. Such as agricultural sector, because of natural disasters limited fertility of land, outdated use of instruments etc.

PART-II

- Q 1. What is the scale of production what are its determinants.
- Q 2. What is the difference between large scale of production and small scale production what are the merits and demerits.
- Q 3. What is the difference between external and internal economies.
- Q 4. Explain the law of increasing return with the help of diagram and schedule also give it's application.
- Q 5. Explain the law of constant returns with the help of diagram and schedule also give it's application.
- Q 6. Explain the law of diminishing returns with the help of diagram and schedule brings out it's application also.
- Q 7. Why does laws of production are called laws of costs. Explain with the help of schedule and diagram.
- Q 8. Explain the law of variable proportions.

